Quantum-Inspired Feature Selection with Hybrid HOG-LBP Features for Facial Expression Recognition

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Abstract

Facial Expression Recognition (FER) is becoming an important part of modern smart systems, helping machines understand human emotions in applications such as healthcare monitoring, human–computer interaction, and surveillance. While deep learning methods have shown strong performance in FER, they often rely on large datasets and high computational power, making them less suitable for resource-limited environments. In this work, we propose a hybrid approach that combines handcrafted feature extraction with quantum-inspired optimization and deep learning for efficient classification. Histogram of Oriented Gradients (HOG) and Local Binary Patterns (LBP) are used to capture important structural and texture details of facial images, while a Quantum-Inspired Genetic Algorithm (QIGA) is applied to select the most relevant features and remove redundancy. The selected features are further refined using a Top-K strategy and classified using an improved Deep Neural Network (DNN). Experiments on the CK+ dataset show that the proposed method achieves a test accuracy of **96.44%**,representing an improvement of approximately **2–3%** over the baseline approach, demonstrating that combining handcrafted features with quantum-inspired optimization can enhance performance while maintaining computational efficiency.
Keywords: Facial Expression Recognition, HOG, LBP, Quantum-Inspired Genetic Algorithm (QIGA), Feature Selection, Deep Neural Network (DNN), CK+ Dataset, Hybrid Model

1 Introduction

Facial expressions are a powerful way to talk to people without using words. They let people say what they want, how they feel, and how they feel mentally. They are very important for how people act and talk to each other in public[17]. Automatic facial expression recognition has emerged as a prominent research area, with applications in emotion-aware systems, intelligent tutoring, driver monitoring, human-computer interaction, surveillance, and mental health assessment [18]. Accurate Facial Expression Recognition (FER) systems enable more natural and intuitive interactions between machines and humans[19].

Traditional FER systems mainly use hand-crafted methods to get features from data. Some of these are Histogram of Orientated Gradients (HOG) [4] and Local Binary Patterns (LBP) [5]. HOG captures information about edges and gradients, which is a good way to show the structural features of facial expressions. LBP encodes local texture patterns and is resilient to variations in illumination [11][16]. These methods are easy to understand and don't use a lot of computer power. Despite their simplicity, HOG and LBP have demonstrated reliable performance in controlled environments, making them a foundational component in the early development of FER systems. They work well for systems and applications that need to work in real time and don't have a lot of computing power.

However, handcrafted feature-based methods often make feature vectors with too many dimensions and features that aren't useful. This can make the math harder and slow down the classification process. So, feature selection methods are necessary to find the most important features and improve overall efficiency [13]. Also, having features that are noisy or not useful can make it harder for classifiers to tell the difference between things. That's why robust feature optimisation is an important part of FER systems.Furthermore, efficient dimensionality reduction not only accelerates processing but also helps in improving model generalization by reducing the risk of overfitting.

However, handcrafted feature-based methods often produce high-dimensional feature vectors with unnecessary and less useful features. This can make the calculations more complicated and hurt the performance of the classification. Feature selection techniques are thus indispensable for discerning the most pertinent features and enhancing overall efficiency [13]. Also, having features that are noisy or not useful can make classifiers less able to tell the difference between things, so robust feature optimisation is an important part of FER systems.

To address these constraints, hybrid methodologies that integrate handcrafted features with deep learning have been investigated. These methods use the learning power of neural networks while taking advantage of the efficiency and interpretability of traditional features [12]. These hybrid frameworks try to find a balance between performance and cost, but having extra features in combined representations is still a big problem. This shows how important it is to have good strategies for optimising features[22]. Additionally, selecting an optimal subset of features is crucial to fully utilize the strengths of hybrid models while avoiding unnecessary computational overhead.

Optimisation-based feature selection methods, including Genetic Algorithms and other metaheuristic approaches, have been widely used to improve feature selection [14]. While these methods provide better exploration compared to conventional techniques, they may suffer from premature convergence and limited diversity in the search process, which can restrict their effectiveness in high-dimensional problems.

Quantum-inspired optimisation algorithms have recently gained popularity because they use probability to represent things and can search more efficiently. Unlike classical optimisation methods, they maintain a superposition of multiple candidate solutions, enabling a more diverse and global search of the solution space. Quantum-Inspired Genetic Algorithms (QIGA) enable enhanced exploration of the feature space and exhibit more stable convergence behaviour relative to traditional methods [6]. This makes them especially good for tasks with a lot of dimensions, like recognising facial expressions, where finding discriminative features is very important for performance.

In this paper, a hybrid FER framework is proposed that integrates handcrafted feature extraction, quantum-inspired feature selection, and deep learning-based classification.

Specifically, HOG and LBP are used to extract complementary structural and texture features from facial images. These features are then optimised using QIGA to eliminate redundancy and retain the most informative subset. A Top-K feature selection strategy is further applied to reduce dimensionality, and the resulting features are used to train an improved Deep Neural Network (DNN) for classification.
The proposed approach aims to achieve high accuracy while maintaining computational efficiency, making it suitable for real-world applications. It is specifically designed to handle high-dimensional feature spaces efficiently while preserving the most discriminative information required for accurate classification. Additionally, the integration of optimized feature subsets helps in reducing redundancy and improving the overall robustness of the system [1]. Experimental evaluation on the CK+ dataset demonstrates that the proposed method outperforms traditional and hybrid approaches, validating the effectiveness of integrating quantum-inspired optimisation with feature-based and deep learning techniques
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2 Related Work

Facial Expression Recognition (FER) has been extensively studied using handcrafted feature extraction techniques, which rely on manually designed features to capture important facial characteristics. Among the most widely used are Histogram of Oriented Gradients (HOG) [4] and Local Binary Patterns (LBP) [5]. HOG focuses on edge directions and structural information to effectively represent facial shapes and contours, while LBP encodes local texture information and is robust to illumination variations [11]. Together, these methods provide a complementary representation by capturing both global structural patterns and fine-grained local textures of facial expressions [16].
Handcrafted features are computationally efficient and interpretable, making them suitable for real-time applications and systems with limited computational resources. However, these approaches have limitations: their performance can degrade under varying lighting conditions, pose variations, and occlusions. Extracted feature vectors are often high-dimensional and may contain redundant or irrelevant information, which negatively impacts classification performance and increases computational complexity [13].
With advances in machine learning, deep learning approaches—particularly Convolutional Neural Networks (CNNs)—have significantly improved FER performance[10]. CNNs automatically learn hierarchical feature representations from raw images, eliminating the need for manual feature design [9]. These models effectively handle variations in lighting, pose, and partial occlusions, making them suitable for complex real-world scenarios. Recent developments, including deeper architectures and attention-based mechanisms, have further enhanced the ability of deep learning models to capture subtle facial variations [8]. However, these methods require large-scale annotated datasets and high computational resources. With limited data, such as the CK+ dataset [2], deep learning models are prone to overfitting and may not generalise well[3].
To address high-dimensional feature spaces, optimisation-based feature selection techniques have been explored. Methods such as Genetic Algorithms and other metaheuristic approaches are widely used to select the most relevant features and improve classification performance [14]. These approaches help reduce redundancy, enhance model efficiency, and improve learning speed. Despite their advantages, traditional optimisation techniques may suffer from issues like premature convergence and limited exploration of the

search space, preventing the model from finding the optimal feature subset in complex, high-dimensional problems.

Recently, quantum-inspired optimisation algorithms have emerged as promising alternatives for feature selection tasks. Quantum-Inspired Genetic Algorithms (QIGA) utilise probabilistic representations based on quantum bits, enabling exploration of multiple solutions simultaneously [6]. This leads to improved exploration and better convergence compared to classical optimisation techniques. Such properties make QIGA particularly suitable for FER, where identifying a compact and discriminative feature subset is crucial for high performance.

To leverage the strengths of both handcrafted features and deep learning, hybrid approaches have been proposed. These methods combine feature extraction techniques such as HOG and LBP with deep neural networks to improve recognition performance [12]. By integrating complementary feature representations, hybrid models aim for better accuracy while maintaining computational efficiency. However, even in hybrid frameworks, redundant features remain a challenge, making effective feature selection essential to fully utilise these models' potential.

Table 1: Comparison of Facial Expression Recognition Methods

Author	**Year**	**Dataset**	**Preprocessing Techniques**	**Acc.**
[1]Askri et al.	2023	CK+	Feature optimization	93.3%
[3]Lucey et al.	2010	CK+	Image alignment	83.3%
[12]Ding et al.	2019	CK+	–	94.0%
[14]Yang	2014	–	Metaheuristic methods	–
[16]Zhao et al.	2007	CK+	Dynamic LBP	91.0%
[19]Chang et al.	2018	CK+	CNN features	95.0%
[21]Koelstra et al.	2012	DEAP	Multimodal processing	89.0%
[23]Jung et al.	2015	CK+	–	92.5%
[24]Zhao et al.	2015	CK+	HOG features	95.8%
[26]Zhang et al.	2022	CK+	–	95.2%
[30]Moore et al.	2011	Multi-view	LBP features	91.5%
Proposed Method	–	CK+	HOG + LBP + Normalization	**96.44%**

Although significant progress has been made in FER, limited work has explored the integration of handcrafted feature extraction, quantum-inspired optimisation, and deep learning within a unified framework. Existing methods typically focus on feature extraction, classification, or optimisation independently. This gap motivates the proposed approach, which combines HOG and LBP features with QIGA-based feature selection and deep learning classification to achieve high accuracy, improved efficiency, and robust performance under varying conditions.

3 Proposed Methodology

The proposed framework integrates handcrafted feature extraction, quantum-inspired optimisation, and deep learning classification with the objective of maximising classification accuracy while reducing feature dimensionality.

This hybrid approach ensures a balance between high accuracy and computational efficiency.

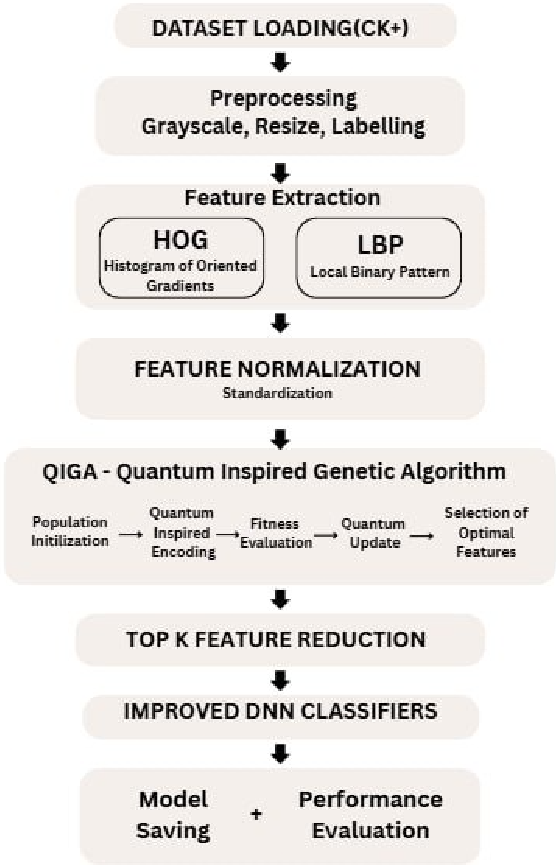

Figure 1: Overview of the proposed facial expressionrecognition framework.

3.1 Feature Representation

Let an input image be represented as $I \in R^{48\times48}$. Two complementary feature descriptors are extracted:

HOG Features: HOG computes gradient orientation histograms over localised regions, capturing structural information such as edges and contours. The image is divided into cells of size 8×8 pixels, each representing a small spatial region for computing gradient orientations. These cells are grouped into blocks of size 2×2 to normalise contrast variations, capturing local shape information while maintaining robustness to illumination changes.

LBP Features: Local Binary Pattern (LBP) is a texture descriptor used for feature extraction in image processing.LBP encodes local texture by comparing each pixel with its neighbouring pixels:

$$LBP(x_c, y_c) = \sum_{p=0}^{P-1} s(i_p - i_c) \cdot 2^p \tag{1}$$

where $LBP(x_c, y_c)$ represents the LBP value at the central pixel located at (x_c, y_c), P denotes the number of neighboring pixels, i_c is the intensity of the central pixel, i_p represents the intensity of the neighboring pixel, and $s(\cdot)$ is a sign function defined as:

$$s(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases} \tag{2}$$

The LBP operator works by comparing each neighboring pixel with the central pixel. If the neighbor's intensity is greater than or equal to the center, it is assigned a value of 1; otherwise, 0. These binary values are then weighted and summed to form a unique LBP code, which captures local texture patterns.
The final feature vector is defined as:

$$F = [F_{HOG} \mid\mid F_{LBP}] \tag{3}$$

where F represents the final feature vector, F_{HOG} denotes the feature vector extracted using Histogram of Oriented Gradients (HOG), and F_{LBP} represents the feature vector obtained using Local Binary Pattern (LBP). The operator $||$ denotes the concatenation of two feature vectors.
The combined feature vector integrates gradient-based shape information from HOG with texture-based information from LBP, thereby enhancing the discriminative power of the feature representation for classification tasks.

3.2 Feature Normalization

Feature normalization is performed to scale the extracted features into a common range, ensuring that no single feature dominates the learning process due to differences in magnitude.The feature vector F is normalised using standard scaling:

$$F' = \frac{F - \mu}{\sigma} \tag{4}$$

where F' represents the normalized feature vector, F denotes the original feature vector, μ is the mean of the feature values, and σ is the standard deviation.
This normalization process transforms the feature distribution to have zero mean and unit variance, which improves the stability and convergence speed of machine learning models.

3.3 Quantum-Inspired Feature Selection

Each candidate solution is represented as a Q-bit vector:

$$Q = [q_1, q_2, ..., q_n], \quad q_i \in [0, 1] \tag{5}$$

where q_i is the probability of selecting the i-th feature. Higher q_i means higher likelihood of selection.
Population size: 20, Generations: 5
Binary solutions are generated via probabilistic collapse:

$$b_i = \begin{cases} 1, & \text{if } rand < q_i \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

where b_i represents the binary state of the i-th feature, q_i denotes the probability associated with the i-th feature, and $rand$ is a random number uniformly distributed in the range $[0, 1]$. This process ensures stochastic selection of features based on their probabilities.

Fitness function (classification accuracy):

$$Fitness(b) = Acc(X_b, y) \tag{7}$$

where $Fitness(b)$ represents the fitness value of the binary solution b, $Acc(\cdot)$ denotes the classification accuracy function, X_b is the subset of features selected according to the binary vector b, and y represents the corresponding class labels.
A Random Forest (RF) classifier with 50 estimators is employed to evaluate the selected feature subsets. The ensemble nature of RF improves classification performance by combining multiple decision trees and reducing overfitting.
The Q-bit update rule is used to iteratively update the probability of feature selection:

$$q_i^{t+1} = q_i^t + \alpha(b_i^{best} - q_i^t) \tag{8}$$

where q_i^t represents the probability of selecting the i-th feature at iteration t, q_i^{t+1} is the updated probability, b_i^{best} denotes the best binary solution obtained so far for the i-th feature, and α is the learning rate controlling the update step size.
In this study, the learning rate is set to $\alpha = 0.05$. This update mechanism gradually adjusts the probabilities towards the optimal binary solution, enabling efficient exploration and exploitation of the feature space.

3.4 Feature Reduction

After the optimization process, the top K most relevant features are selected to form the final feature vector:

$$F_{final} \in R^K, \quad K = 500 \tag{9}$$

where F_{final} represents the final selected feature vector, R^K denotes a K-dimensional real-valued feature space, and K is the number of selected features.
This selection process significantly reduces the dimensionality of the feature space while retaining the most discriminative features. As a result, it improves computational efficiency, reduces model complexity, and enhances classification performance.

3.5 Classification Using DNN

The final feature vector is input to a deep neural network with multiple fully connected layers:

$$h_1 = ReLU(W_1 F_{final} + b_1), \quad \text{(512 neurons)} \tag{10}$$

$$h_2 = ReLU(W_2 h_1 + b_2), \quad \text{(256 neurons)} \tag{11}$$

$$h_3 = ReLU(W_3 h_2 + b_3), \quad \text{(128 neurons)} \tag{12}$$

The number of neurons in each layer controls the model's capacity to learn complex patterns, with deeper layers capturing higher-level abstractions.
The output layer uses Softmax for classification:

$$\hat{y} = Softmax(W_4 h_3 + b_4) \tag{13}$$

The model is trained with the Adam optimiser (learning rate 0.001) for 20 epochs. Dropout layers (rates 0.4 and 0.3) and batch normalisation are used to prevent overfitting and stabilise convergence..

4 Results

The proposed model is evaluated on the Extended Cohn-Kanade (CK+) dataset, which is a widely used benchmark dataset for facial expression recognition. The dataset contains facial images of subjects displaying different emotions, captured under controlled conditions. It includes a total of 981 image sequences from 123 subjects, with peak expression frames commonly used for classification tasks.

For this study, the dataset is organised into seven emotion categories, namely anger, disgust, fear, happiness, sadness, surprise, and neutral. Each image is converted to grayscale and resized to a fixed resolution of 48×48 pixels to ensure uniformity across the dataset.

After preprocessing, the dataset consists of approximately 981 labelled facial images. The dataset is split into training and testing sets using an 80:20 ratio, where 80% of the data is used for training the model, and the remaining 20% is reserved for testing. This split ensures that the model is evaluated on unseen data, providing a reliable estimate of its generalisation performance.

The model is trained for 20 epochs using the Adam optimiser with a learning rate of 0.001. During training, the model learns to map the optimised feature vectors to their corresponding emotion labels. The performance of the model is evaluated using standard metrics such as accuracy, precision, recall, and F1-score.

Table 2: Configuration of QIGA and DNN Parameters

Parameter	**Value**
Image Size	48×48
Feature Extraction	HOG + LBP
HOG Cell Size	8×8
LBP Parameters (P, R)	(8, 1)
Total Features (Before Selection)	High-dimensional
Feature Selection Method	QIGA
Population Size	20
Generations	5
Learning Rate (QIGA)	0.05
Selected Features (Top-K)	500
Classifier	Deep Neural Network
Optimizer	Adam
Learning Rate (DNN)	0.001
Epochs	20
Batch Size	Full Batch Training

4.1 Overall Performance

The suggested model gets a test accuracy of 96.44%, which shows that it can generalise well. The high accuracy shows that combining handcrafted feature extraction with quantum-inspired feature selection and deep learning classification works well.

4.2 Class-wise Performance Analysis

The classification report shows that most emotion classes have consistently high precision, recall, and F1-score. This means that the model isn't biased toward any one class and keeps its performance balanced.
There are small differences in performance when expressions have small visual differences, like fear and surprise. These differences are to be expected because the facial features overlap, which makes it hard for even the most advanced FER systems.

4.3 Confusion Matrix Analysis

From the confusion matrix, it can be observed that the majority of samples are correctly classified, as indicated by the high values along the diagonal. This demonstrates the model's strong ability to distinguish between different facial expressions.
However, a few misclassifications are present between certain emotion classes, particularly those with similar visual characteristics. For instance, minor confusion can be observed between closely related expressions such as fear and surprise, which share overlapping facial features.

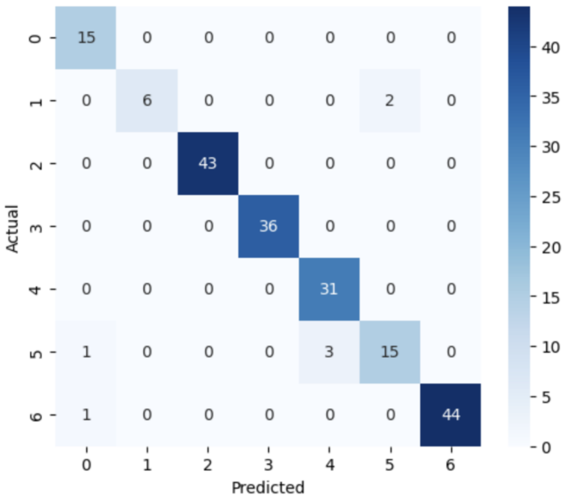

Figure 2: confusion matrix

Overall, the confusion matrix highlights that the proposed model achieves high classification accuracy with minimal errors, while the observed misclassifications are consistent with challenges commonly reported in facial emotion recognition tasks.

4.4 Effect of Feature Selection

The integration of QIGA is very important for improving performance. The algorithm cuts down on noise and redundancy in the feature space by only choosing the most useful features..
Compared to using the full feature set, QIGA-based selection results in:

- Improved classification accuracy
- Reduced computational complexity

- Faster model convergence

This shows that choosing the right features is very important for getting good results, especially when using hand-crafted features.

4.5 Comparison with Other Models

To better evaluate the effectiveness of the proposed approach, it is compared with existing hybrid methods that combine feature extraction and machine learning techniques.

Table 3: Comparison with Existing and Base FER Models

Method	**Feature Strategy**	**Classifier**	**Accuracy**
HOG + LBP + SVM	Handcrafted Features	SVM	90.2%
LBP + CNN Hybrid	Texture + Deep Features	CNN	92.8%
HOG + CNN Hybrid	Gradient + Deep Features	CNN	93.5%
GA-based Feature Selection + DNN	Optimized Features	DNN	94.1%
Base Paper (QGOA + DNN)	Optimized Deep Features	DNN	93.3%
Proposed Method (QIGA + DNN)	Hybrid + Quantum Optimization	DNN	**96.44%**

It can be observed that the proposed method outperforms both traditional and hybrid approaches, as well as the base model, achieving the highest accuracy. The improvement highlights the effectiveness of integrating handcrafted features with quantum-inspired feature selection and deep learning.

4.6 Detailed Analysis of Experimental Results

This paper introduces a hybrid facial expression recognition framework that combines handcrafted feature extraction, quantum-inspired feature selection, and deep learning classification. The suggested method works well, as shown by its 96.44The results show that using both old and new optimisation methods together can lead to better performance with less computing power. The next step will be to apply the proposed framework to bigger datasets like FER2013 and RAF-DB. Adding attention mechanisms and looking into real quantum computing methods could also make things work better.HOG gives it information about the structure of the face, and LBP gives it information about the texture of the face. This strengthens the model.
The optimised feature subset helps by getting rid of features that aren't useful. The deep neural network can focus on the important features, which makes the model work better and faster. The suggested method is better than ones that mix different features. It is more precise. Works better. This indicates that employing quantum-inspired optimisation with feature descriptors is advantageous. The suggested framework uses quantum-inspired optimisation and hybrid feature extraction to help recognise facial expressions.

5 Conclusion and Future Work

This paper introduces a hybrid facial expression recognition framework that combines handcrafted feature extraction, quantum-inspired feature selection, and deep learning

classification. The suggested method works well, as shown by its 96.44% accuracy on the CK+ dataset.
The results show that using both old and new optimisation methods together can lead to better performance with less computing power. The next step will be to apply the proposed framework to bigger datasets like FER2013 and RAF-DB. Adding attention mechanisms and looking into real quantum computing methods could also make things work better.

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